

Let's consider a 6×6 grayscale image and go step by step through the Harris corner detection process.

Step 1: Consider a 6×6 Image

$$I = \begin{bmatrix} 10 & 12 & 15 & 18 & 20 & 22 \\ 14 & 16 & 19 & 22 & 24 & 26 \\ 20 & 22 & 25 & 28 & 30 & 32 \\ 26 & 28 & 31 & 34 & 36 & 38 \\ 30 & 32 & 35 & 38 & 40 & 42 \\ 34 & 36 & 39 & 42 & 44 & 46 \end{bmatrix}$$

Step 2: Compute Gradients I_x and I_y using Sobel Filters

We use the Sobel operators:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Applying convolution to get horizontal gradient I_x :

$$I_x = \begin{bmatrix} 4 & 6 & 6 & 6 & 6 & 4 \\ 6 & 8 & 8 & 8 & 8 & 6 \\ 10 & 12 & 12 & 12 & 12 & 10 \\ 10 & 12 & 12 & 12 & 12 & 10 \\ 10 & 12 & 12 & 12 & 12 & 10 \\ 8 & 10 & 10 & 10 & 10 & 8 \end{bmatrix}$$

And vertical gradient I_y :

$$I_y = \begin{bmatrix} 4 & 6 & 10 & 10 & 10 & 8 \\ 6 & 8 & 12 & 12 & 12 & 10 \\ 6 & 8 & 12 & 12 & 12 & 10 \\ 6 & 8 & 12 & 12 & 12 & 10 \\ 6 & 8 & 12 & 12 & 12 & 10 \\ 4 & 6 & 10 & 10 & 10 & 8 \end{bmatrix}$$

Step 3: Compute Products of Gradients

$$I_x^2 = \begin{bmatrix} 16 & 36 & 36 & 36 & 36 & 16 \\ 36 & 64 & 64 & 64 & 64 & 36 \\ 100 & 144 & 144 & 144 & 144 & 100 \\ 100 & 144 & 144 & 144 & 144 & 100 \\ 100 & 144 & 144 & 144 & 144 & 100 \\ 64 & 100 & 100 & 100 & 100 & 64 \end{bmatrix}$$

$$I_y^2 = \begin{bmatrix} 16 & 36 & 100 & 100 & 100 & 64 \\ 36 & 64 & 144 & 144 & 144 & 100 \\ 36 & 64 & 144 & 144 & 144 & 100 \\ 36 & 64 & 144 & 144 & 144 & 100 \\ 36 & 64 & 144 & 144 & 144 & 100 \\ 16 & 36 & 100 & 100 & 100 & 64 \end{bmatrix}$$

$$I_x I_y = \begin{bmatrix} 16 & 36 & 60 & 60 & 60 & 32 \\ 36 & 64 & 96 & 96 & 96 & 60 \\ 60 & 96 & 144 & 144 & 144 & 100 \\ 60 & 96 & 144 & 144 & 144 & 100 \\ 60 & 96 & 144 & 144 & 144 & 100 \\ 32 & 60 & 100 & 100 & 100 & 80 \end{bmatrix}$$

Step 4: Compute Harris Matrix M for a window

We take a 3×3 region centered at pixel (3,3) and compute:

$$M = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix}$$

Using a Gaussian-weighted sum, assume:

Using a Gaussian-weighted sum, assume:

$$M = \begin{bmatrix} 900 & 800 \\ 800 & 960 \end{bmatrix}$$

Step 5: Compute Harris Response R

$$R = \det(M) - k \cdot (\text{trace}(M))^2$$

Using $k = 0.04$:

$$\det(M) = (900)(960) - (800)(800) = 864000 - 640000 = 224000$$

$$\text{trace}(M) = 900 + 960 = 1860$$

$$R = 224000 - 0.04(1860^2) = 224000 - 138297.6 = 85702.4$$

Step 6: Thresholding

If $R > T$, where $T = 50000$, we classify the pixel as a corner. Since $85702.4 > 50000$, we detect a corner.

$$I = \begin{bmatrix} 10 & 10 & 10 & 10 & 10 & 10 \\ 10 & 20 & 20 & 20 & 20 & 10 \\ 10 & 20 & 80 & 80 & 20 & 10 \\ 10 & 20 & 80 & 80 & 20 & 10 \\ 10 & 20 & 20 & 20 & 20 & 10 \\ 10 & 10 & 10 & 10 & 10 & 10 \end{bmatrix}$$

1. Gaussian Smoothing

$$G = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

→ Example patch.

$$\begin{bmatrix} 20 & 20 & 20 \\ 20 & 80 & 80 \\ 20 & 80 & 80 \end{bmatrix}$$

$$\frac{1}{16} [1 \cdot 20 + 2 \times 20 + 1 \times 20 + \dots] = \frac{860}{16} = 53.75$$

$$I_s = \begin{bmatrix} 16 & 23 & 23 & 16 \\ 23 & 54 & 54 & 23 \\ 23 & 54 & 54 & 23 \\ 16 & 23 & 23 & 16 \end{bmatrix}$$

2. Sobel gradients

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$P_x = \begin{bmatrix} 16 & 23 & 23 \\ 23 & 54 & 54 \\ 23 & 54 & 54 \end{bmatrix}$$

$$G_x = -1 \times 16 + 23 \times 0 + 1 \times 23 + \dots = 100$$

$$G_y = -100$$

$$G = G_x + j G_y = 100 - j100$$

$$|G| = \sqrt{100^2 + 100^2} = 141.4$$

$$\theta = \tan^{-1}\left(\frac{-100}{100}\right) = -45$$

$$= 180 - 45 = 135$$

$$G = \begin{vmatrix} 141 & 141 \\ 141 & 141 \end{vmatrix} \xrightarrow{\text{why 4}} \begin{vmatrix} 16 & 23 & 23 & 16 \\ 23 & 54 & 54 & 23 \\ 23 & 54 & 54 & 23 \\ 16 & 23 & 23 & 16 \end{vmatrix}$$

Non maxima suppression

Q $|u|$ at $(0,0) > |u|$ at $(0,1)$

A = NO suppress it. Do same for other diagonal as diagonal represent

135° Hence $G = \begin{vmatrix} 0 & 141 \\ 141 & 0 \end{vmatrix}$

$$T_H = 120, T_L = 80$$

$$E = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

angle usage

0° Horizontal

45° Diagonal

90° Vertical

135° Diagonal

Pixels Coordinates to check

$(x+1, y) (x-1, y)$

$(x+1, y-1) (x-1, y+1)$

$(x, y+1) (x, y-1)$

$(x-1, y-1) (x+1, y+1)$

NMS non-max suppression

(3)

G_2

0	0	40	0	0	0
0	60	110	65	30	0
0	55	180	75	45	0
0	20	70	120	60	0
0	0	35	50	25	0
0	0	0	0	0	0

$T_H = 100$, $T_L = 50$. Strong ≥ 100
Weak $50 < \text{value} < 100$

Non-edge < 50

~~00 55 00 0~~
~~060 110 65~~

0	0	0	0	0	0
0	60	110	65	0	0
0	55	180	75	0	0
0	0	70	120	60	0
0	0	0	50	0	0
0	0	0	0	0	0

* 60, 65, 55, 75
are connected
to 110 &
keep.

0 0 40 0 0 (55) → weak reminder
65 0

SIFT descriptor

I_1



Keypoint localization
and refinement

K_{A1}

K_{A1}

K_{A1}

I_2



Keypoint localization
refinement

K_{B1}

K_{B2}

K_{B3}

5 — Best match point

12

8 — Second best

Lowe's ratio

$$\frac{\text{Best match}}{\text{Second best}} < 0.75$$

$$\frac{5}{8} \leq 0.625 < 0.75$$

We accept the match that best
will be considered for feature matching. Ratio
just validates the match.

Keypoint localization

1. Scale-space

2. Keypoint

Extrema detection
localization

3. Orientation
assignment
4. Descriptor

1. Gaussian blur + Dog + Key point detection
2. Taylor Expansion + Hessian matrix (key point refinement) ⁽²⁾
3. Gradient magnitude + Histogram
4. 16x16 region $\rightarrow$ 128 D Vector
5. Euclidean distance + Lowe's ratio

I At page
4 $\rightarrow$ 5x5 ($w=1, 1.6, 2.0$)

$w=1$ $\frac{1}{273}$ $\begin{bmatrix} 1 & 4 & 7 & 4 & 1 \\ 4 & 16 & 26 & 16 & 4 \\ 7 & 26 & 41 & 26 & 7 \\ 4 & 16 & 26 & 16 & 4 \\ 1 & 4 & 7 & 4 & 1 \end{bmatrix}$ $w=1.6$ $\frac{1}{837}$ $\begin{bmatrix} 3 & 7 & 11 & 7 & 3 \\ 7 & 21 & 36 & 21 & 7 \\ 11 & 36 & 61 & 36 & 11 \\ 7 & 21 & 36 & 21 & 7 \\ 3 & 7 & 11 & 7 & 3 \end{bmatrix}$

Physically $w=1 \rightarrow 3 \times 3 \rightarrow$ approx by 3×3
 $w=1.6 \rightarrow 5 \times 5$
 $w=2.0 \rightarrow 13 \times 13 \rightarrow 5 \times 5$

Generally $2 \times 3w + 1 = \text{size of kernel}$
 $1.6 - 1 \rightarrow \text{DOG1}$
 $2 - 1.6 \rightarrow \text{DOG2}$

I $\rightarrow$ $I(w=1)$
 $I(w=1.6)$
 $I(w=2.0)$

27 neighbour Comparison $\begin{matrix} w=1.0 & \rightarrow & 9 \text{ neighbours} \\ w=1.6 & \times & \rightarrow & 8 \text{ neighbours} \\ w=2.0 & \rightarrow & 9 \text{ neighbours} \end{matrix}$

if value < all neighbours

Key point -

$$x = -H^{-1} \nabla D$$

$$\nabla D = \begin{bmatrix} 0 \\ -0.5 \end{bmatrix}$$

$$H = \begin{bmatrix} 6 & 0.5 \\ 0.5 & 5 \end{bmatrix}$$

x, y	$G(x_1)$	$G(x_2)$	D
			-5
9, 14	150	145	-7
10, 14	160	153	-6
11, 14	158	152	-7
9, 15	162	155	-10
* 10, 15	170	160	-7
11, 15	165	158	-7
9, 16	157	150	-8
10, 16	168	160	-6
11, 16	160	154	

at (10, 15) $D_{old} = -10$

$$\nabla D_x = \frac{\nabla D}{\partial x} = \frac{D(11, 15) - D(9, 15)}{2} = \frac{-7 - (-10)}{2} = 0$$

$$\nabla D_y = \frac{\nabla D}{\partial y} = \frac{D(10, 16) - D(10, 14)}{2} = \frac{-6 - (-7)}{2} = 0.5$$

$$\frac{d^2 D}{dx^2} = D(11, 15) - 2D(10, 15) + D(9, 15) = -7 + 2(-10) - (-7) = -10$$

$$H^{-1} = \frac{1}{\det(H)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \begin{bmatrix} 0.168 & -0.017 \\ -0.017 & 0.202 \end{bmatrix}$$

$$x_A = x_0 - H^{-1} \nabla D = \begin{bmatrix} 10 \\ 15 \end{bmatrix} - \begin{bmatrix} \quad \\ \quad \end{bmatrix} \begin{bmatrix} 0 \\ -0.5 \end{bmatrix}$$

$$I = \begin{bmatrix} 50 & 80 & 90 & 60 & 30 \\ 70 & 100 & 110 & 80 & 40 \\ 90 & 120 & 130 & 100 & 60 \\ 70 & 100 & 110 & 80 & 40 \\ 50 & 80 & 90 & 60 & 30 \end{bmatrix}$$

(100) keypoint
Build a 16x16 window
around keypoint

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$I_{G_x} = ?$$

$$I_{G_y} = ?$$

$$M = \sqrt{I_{G_x}^2 + I_{G_y}^2}$$

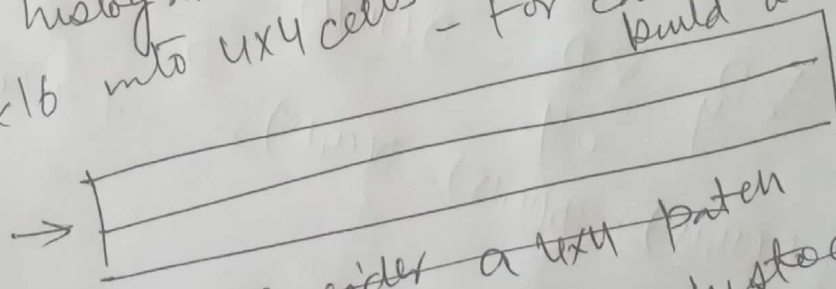
$$\Rightarrow \theta = \tan^{-1} \left(\frac{I_{G_y}}{I_{G_x}} \right)$$

at each pixel.

Build histogram (0-360) at interval of 10-45
8 bins

Divide 16x16 into 4x4 cells - For each cell of 4x4 build a histogram of 8 bins

Noting is performed for dominant direction



Consider a 4x4 patch
Hence 16 such histograms are build resulting in 16x8=128 D vectors

128D vector making each histogram bin
is normalized (5)

$$D' = \frac{D}{\|D\|}$$

D is vector of gradient magnitudes from all sub regions

After normalization any value > 0.2 is clamped
and vector is renormalized.

Now these normalized from I_A and I_B are
compared using Lowe's test (Page 4)